

SLVP: Every Root of a Polynomial

GFDL-1.3-or-later

SLVP takes the coefficients of a polynomial of any degree from a vector in **X** and returns every root, real and complex, in one keystroke. It completes a family: SLVQ solves the quadratic and SLVC the cubic, both from stack registers, and until now degree four was where the calculator stopped. HP 50g owners will recognise the shape of the function as PROOT, which has been the workhorse of control engineers and filter designers on that machine for twenty years.

The first half of this note is about using the function: what it takes, what it returns, and six worked examples ending with an internal rate of return. The second half is about what happens behind the keystroke: the companion matrix, the QR eigensolver that C47 already carries, what accuracy to expect and where the limits are.

Numerical results in this document are of three kinds. Results computed by hand are given with their working, so the reader can check them. Results the C47 test suite asserts are identified by the file that asserts them, `src/testSuite/tests/slv.p.txt`, which holds thirty-three cases covering every path through the function. Independent confirmations were computed with 60-digit decimal arithmetic outside the calculator and are marked as such. Nothing in this note has to be read off real hardware.

WARNING

SLVP ships on the DM42n and the R47, and in the simulator. It is not in any DM42 package: the DM42 flash images are full to within a kilobyte, and SLVP needs the eigensolver (OPTION_EIGEN) beneath it. On a DM42 the catalog shows SLVP struck through, which is the calculator's way of saying a function exists in the family but not in this build.

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REV	DATE	DESCRIPTION
0.1	2026-07-30	First draft, written alongside the implementation.

1 The Coefficient Vector

A polynomial is handed to SLVP as a vector of its coefficients, highest degree first, the same order in which the polynomial is written on paper. Every power must have its slot, so a polynomial with no x^2 term carries an explicit 0 in the x^2 position:

$$2x^3 - 3x + 5 = 2x^3 + 0x^2 - 3x + 5 \implies [2, 0, -3, 5]$$

The vector may be a row or a column, and its elements may be real or complex. A vector of m elements describes a polynomial of degree $m - 1$, and SLVP returns exactly $m - 1$ roots, always.

Two zero patterns are handled the way algebra handles them. Leading zeros carry no degree, so $[0, 0, 1, -3, 2]$ is simply the quadratic $x^2 - 3x + 2$: they are dropped silently, which is the same convention SLVC uses when its leading coefficient is zero. Trailing zeros each contribute one exact root at 0, so $x^4 - 3x^3 + 2x^2$ returns $[2, 1, 0, 0]$ without the solver ever running on the deflated part.

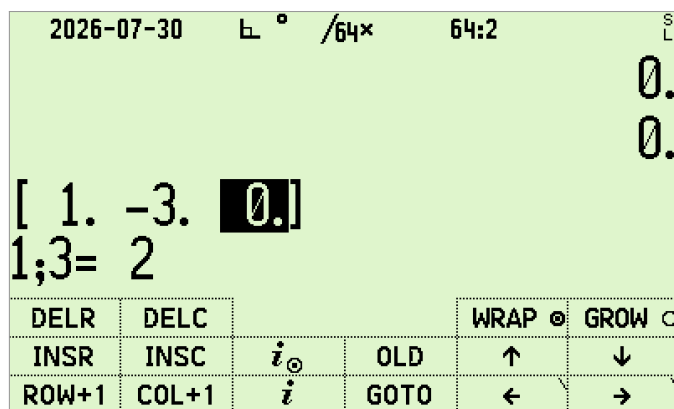
2 Entering the Vector and Running SLVP

All keystrokes in this note are for the C47 keyboard; the maps in the appendix locate every key on both machines. Enter the dimensions, create the matrix, and fill it. For $x^2 - 3x + 2$:

1 [ENTER] 3 [i] [MATX] [NEW] [EDIT] → 1;1= 0

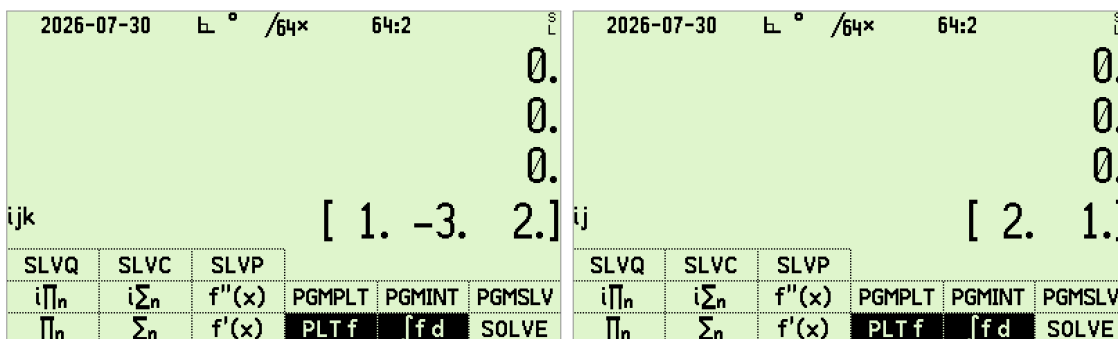
The matrix editor is now open on a 1×3 vector of zeros. Type each coefficient and step right with the [→] soft key:

1 [→] 3 [+/-] [→] 2



The editor with two coefficients stored and the third, 2, being typed into element 1;3. The editor's own menu carries the arrows; WRAP and GROW govern what the cursor does at the last element. The tiny S over L at the top right is the simulator's stack-lift indicator; real hardware carries the battery gauge there.

Close the editor and run the solver. SLVP sits beside its family on the g row of the ADV menu:



Left, the ADV menu with the coefficient vector in **X**: the solver family SLVQ, SLVC, SLVP sits on the g row. Right, the answer: both roots of $x^2 - 3x + 2$, largest magnitude first.

That is the whole of it. The result replaces **X**, the coefficient vector goes to **L** as LASTx always has it, and the rest of the stack is untouched.

NOTE

SLVP is also in the catalog under its own name, is assignable to any key, and is a plain program step: a program that puts a coefficient vector in **X** and executes SLVP gets the root vector back in **X**. Section 6 gives the three-step program this note used to prove it.

2.1 Reading the result

The roots come back as a $1 \times n$ vector, sorted by descending magnitude. If every root is real, the result is a real vector. If any root is complex, the whole vector is complex, the complex-result flag CPXRES is set the same way SLVQ sets it, and conjugate pairs sit next to each other in the ordering. Equal-magnitude roots keep the order the solver found them in; the fifth roots of 32 in Section 3 arrive in angle order for exactly that reason.

3 Worked Examples

3.1 Degree five, integer roots

The polynomial $(x-1)(x-2)(x-3)(x-4)(x-5) = x^5 - 15x^4 + 85x^3 - 225x^2 + 274x - 120$:

[1, -15, 85, -225, 274, -120]  ADV  SLVP → X: [5. 4. 3. 2. 1.]

All five roots to 34 digits. The test suite asserts this case, and it is deliberately the same polynomial the EIGVAL tests have solved as a hand-built companion matrix since the eigensolver landed: SLVP is that construction with the bookkeeping done for you.

3.2 A complex pair from real coefficients

$x^2 - 2x + 5$ has no real roots. SLVP does not need to be told:

[1, -2, 5]  ADV  SLVP → X: [1.+i2. 1.-i2.]

2026-07-30 L ° /64x 64:2 S					
0.					
0.					
0.					
[1.+i×2. 1.-i×2.]					
SLVQ	SLVC	SLVP			
$i\prod_n$	$i\sum_n$	$f''(x)$	PGMPLT	PGMINT	PGMSLV
$\prod_n$	$\sum_n$	$f'(x)$	PLT f	$\int f d$	SOLVE

The conjugate pair $1 \pm 2i$. The result vector is complex, and CPXRES has been set on the way out, so later real operations that stray into the complex plane will follow.

3.3 All the roots of $x^5 = 32$

Ask for the five fifth roots of 32 by solving $x^5 - 32 = 0$:

[1, 0, 0, 0, 0, -32]  ADV  SLVP → X: [2. 0.618...+i1.902... ...]

All five sit on the circle of radius 2, in angle order starting from the real root. The suite asserts every one to 34 digits against the closed forms $2\cos(2\pi k/5) + 2i\sin(2\pi k/5)$, whose real parts are $(\sqrt{5}-1)/2$ and $-(\sqrt{5}+1)/2$. This example takes a different path through the function than the others, and Section 5.2 explains why.

3.4 Complex coefficients

Coefficients may be complex. $x^2 - 3ix - 2$ factors as $(x-i)(x-2i)$:

[1, -3i, -2]  ADV  SLVP → X: [0.+i2. 0.+i1.]

Complex elements are typed in the editor the same way they are typed into the stack, with i between the parts. The suite also asserts $x^4 + ix^3 - 2$, a quartic with complex coefficients whose four roots were confirmed outside the calculator by 60-digit residual evaluation. One of them is exactly i , and it comes back carrying a real part of -4×10^{-40} : for complex coefficients SLVP delivers the solver's raw values, and Section 5.3 says why it must.

3.5 Deflation

$x^4 - 3x^3 + 2x^2$ is $x^2(x-1)(x-2)$:

[1, -3, 2, 0, 0]   → X: [2. 1. 0. 0.]

The two zero roots are exact and cost nothing: the solver runs on the deflated quadratic only.

3.6 Internal rate of return

A polynomial root finder is a rate solver. An investment of 100 returns 30, 40, 50 and 60 over the following four periods. Writing $v = 1/(1+r)$ for the discount factor per period, the net present value is zero when

$$60v^4 + 50v^3 + 40v^2 + 30v - 100 = 0,$$

which is a coefficient vector, latest cash flow first:

[60, 50, 40, 30, -100]   → X: [... 0.8007... ...]

The one positive real root is $v = 0.80071529 \dots$; the other three roots are the complex artefacts every cash-flow polynomial carries, and they are ignored. Then $r = 1/v - 1 = 0.248883 \dots$, an internal rate of return of 24.888 %. Checked by hand: at $r = 24.888 \%$ the four discounted inflows sum to 100.000.

TIP

Sign-change counting says how many positive real roots to expect: one sign change in the cash flow, one positive root, and the IRR is unambiguous. A flow with more sign changes can have several positive roots, and SLVP shows all of them, which is precisely what the single-answer IRR button on a financial calculator cannot do.

4 What SLVP Refuses

Everything SLVP refuses is refused whole: **X**, **L** and the rest of the stack are exactly as they were, so a program can test and retry. The suite asserts each row of Table 1, including that the input survives.

Table 1 Every refusal, its error, and what triggers it.

INPUT	ERROR	WHY
Not a matrix	Invalid data type	A scalar, string or integer in X : the coefficients must be a vector
A matrix, not a vector	Matrix mismatch	Both dimensions exceed 1
Degree 0	Function domain	A lone constant, an all-zero vector, or leading zeros down to a constant: no degree, no roots
Degree beyond memory	RAM full	The QR workspace grows as the square of the degree; see Section 5.4

2026-07-30 L ° /64x 64:2 S L					
0.					
An argument exceeds the function domain					
0.					
[5.]					
SLVQ	SLVC	SLVP			
iΠ _n	iΣ _n	f''(x)	PGMPLT	PGMINT	PGMSLV
Π _n	Σ _n	f'(x)	PLT f	∫ f d	SOLVE

The degree-0 refusal: a 1×1 vector is a constant, and a constant has no roots. The vector is still in **X**, untouched.

5 Behind the Keystroke

5.1 The companion matrix

SLVP contains no root-finding iteration of its own. It normalises the polynomial to monic form at 75-digit precision, builds the companion matrix, and hands it to the shifted Householder-QR eigensolver that EIGVAL has used since it landed, working at the same 75 digits. For the monic polynomial $x^d + b_{d-1}x^{d-1} + \dots + b_0$ the companion matrix is

$$C = \begin{pmatrix} 0 & 1 & & \\ & \ddots & \ddots & \\ & & 0 & 1 \\ -b_0 & -b_1 & \dots & -b_{d-1} \end{pmatrix},$$

and its eigenvalues are exactly the roots. The eigensolver sorts them by descending magnitude and strips conjugate-pair noise, and SLVP rounds the 75-digit results to the 34 digits a register holds. Degrees 1 and 2 inside the engine take closed forms rather than iterating, so small polynomials do not pay for machinery they do not need.

The pipeline in order: validate, drop leading zeros, deflate trailing zeros, normalise to monic, special-case $x^d = w$, otherwise build the companion and iterate, then assemble, sort and type the result. Only after all of that succeeds does anything touch the stack, which is what makes the refusals of Section 4 clean.

5.2 Why $x^d = w$ takes a different path

The companion matrix of $x^d - w$ has ones on the superdiagonal and a single entry in the bottom row, and its eigenvalues all share one magnitude. A shifted QR iteration lives on magnitude gaps to peel eigenvalues off, so this shape is the one polynomial the engine is poor at, and the engine screens it out rather than delivering slow garbage. SLVP therefore solves two-term polynomials in closed form: magnitude $|w|^{1/d}$, angles $(\theta + 2\pi k)/d$, computed at 75 digits. That is why the fifth roots of 32 arrive in angle order rather than sorted: all five magnitudes are equal by construction.

The screen in the engine reads only the real parts of the bottom row. A complex polynomial whose middle coefficients are all purely imaginary while its constant is real, like $x^2 - 3ix - 2$ above, looks circulant to that screen although it is not a two-term polynomial at all. SLVP sidesteps the false positive with a diagonal

similarity transform: the companion is scaled so its superdiagonal is $\frac{1}{2}$ instead of 1, which leaves every eigenvalue exactly where it was and makes the matrix invisible to the screen. The suite asserts both quartic and quadratic cases through this path.

5.3 Accuracy

For well-separated roots the answer is simply correct to the display: the engine carries 75 digits and the register keeps 34. The Wilkinson polynomial of degree 10, whose roots 1, ..., 10 are the classical stress test for coefficient sensitivity, comes back as the exact integers to all 34 digits; the suite asserts it.

Repeated roots are different, and no algorithm escapes it: a root of multiplicity m is determined by the coefficients only to about $75/m$ digits. $(x-1)^4$ returns a tight cluster of four values within 3×10^{-18} of 1, two of them carrying imaginary parts of about 7×10^{-19} , and the result is typed complex because that is what the numbers are. The suite bakes these exact cluster values in, with a comment saying why, so any change to this behaviour will show. Read a tight cluster with tiny imaginary parts as a repeated real root; the cluster diameter tells you the multiplicity's worth of digits you still have.

One cleanup is applied on the way out, and only when every coefficient is real. For a real polynomial a component more than 36 orders of magnitude below its root's dominant component is QR convergence residue, not data: a near-real conjugate pair $a \pm bi$ enters the real coefficients through b^2 , which caps how small a genuine b can be at roughly 17 orders below a . Stripping the residue is what lets $x^2 + 1$ return $\pm i$ with the real parts exactly zero. Complex coefficients carry independent exponents for the two parts, so a genuinely tiny component *is* encodable there, no threshold is safe, and SLVP delivers the solver's raw values: the exact root i of $x^4 + ix^3 - 2$ arrives with a harmless -4×10^{-40} real part, which is the honest answer.

5.4 Degree limits

The QR workspace grows as the square of the degree, in 75-digit reals, taken from calculator memory and returned when done. On the 256-KiB pool of the DM42n and R47 the practical ceiling is around degree 14; beyond it SLVP refuses with RAM FULL before touching the stack, and the suite asserts a degree-20 refusal to pin the behaviour. The two-term closed form of Section 5.2 needs no matrix, so $x^{100} = 1$ is fine. The Wilkinson polynomial of degree 10 solves in about nine hundred QR iterations; a repeated-root cluster drives the iteration to its cap and takes visibly longer, which is the hourglass you may see on a hard polynomial.

5.5 Command reference

Table 2 SLVP at a glance.

TAKES	X : real or complex vector, $1 \times m$ or $m \times 1$, $m \geq 2$, highest-degree coefficient first
LEAVES	X : all n roots as a $1 \times n$ vector, descending magnitude, real if every root is real, else complex
READS	nothing else
WRITES	L (the coefficient vector); CPXRES set when the result is complex
REFUSES	non-matrix, non-vector, degree 0, degree beyond memory; always with the stack untouched (Table 1)

6 SLVP in a Program

SLVP is an ordinary step. The program this note used to prove the claim is three steps long:

```
0001: LBL 'PSLV'
0002:  SLVP
0003:  END
```

Open a new program with `[XEQ][.][.]` and switch to program-entry mode with `[f][PRGM]`. Start the label with `[g][LBL]`, then `[f][α]`, type `[P][S][L][V]`, and `[ENTER]`. SLVP itself goes in from the catalog or from `[f][ADV]` exactly as it is run interactively. Leave program-entry mode with `[f][PRGM]`. With the coefficient vector in **X**, `[XEQ]` PSLV returns the roots, and a wrong input mid-program halts with the same errors as Table 1.

7 Availability and Cost

SLVP is gated by `OPTION_SLVP`, which requires `OPTION_EIGEN` beneath it and is enabled for the DM42n, the R47 and the simulator. All four DM42 packages exclude it; the measured cost of the option on the DM42n build, with the arm-gnu 13.2.Rel1 toolchain, is 2024 bytes of flash and no static RAM at all: every working byte is borrowed from calculator memory for the duration of the keystroke. Two kilobytes for every root of every polynomial is the companion-matrix dividend: the expensive machinery was already on board, tested, and waiting.

The C47 keyboard

Primary function on the key; orange = f (shift once) above; blue = g (shift twice) below.

$\rightarrow I$ $\Sigma+$ a b/c	y^x $1/x$ #	x^2 $\sqrt{x}$.ms	10^x LOG .d	e^x LN LBL	α XEQ GTO
$ x $ STO $\angle$	% RCL $\Delta\%$	π $R\downarrow$ $\sqrt[y]{x}$	ASIN SIN i	ACOS COS $\rightarrow R$	ATAN TAN $\rightarrow P$
COMPLEX ENTER CPX	LASTx $x\leftrightarrow y$ STK	MODE $+/-$ TRG	DISP EEX EXP	UNDO $\leftarrow$ CLR	
BST $\uparrow$ REGS	EQN 7 HOME	ADV 8 FIN	MATX 9 X.FN	STAT $\div$ PLOT	
SST $\downarrow$ FLGS	BASE 4 BITS	CONV 5 CLK	FLAG 6 REAL	PROB $\times$ INTS	
f/g	ASN 1 KEYS	USER 2 α .FN	P.FN 3 LOOP	PRINT $-$ I/O	
OFF EXIT SNAP	VIEW 0 STOPW	SHOW $\cdot$ INFO	PRGM R/S TEST	CAT $+$ CNST	

The R47 keyboard

The R47 has two physical shift keys: a yellow **f** and a blue **g**. Primary on the key; f above, g below.



References

- [1] C47/R47 Application Note 020, *Matrix and Vector*.
- [2] C47/R47 Application Note 022, *RPN, Solve, Integration, Plot*.
- [3] C47/R47 Application Note 029, *First Steps*.
- [4] C47/R47 Application Note 030, *Measurement Uncertainty*.
- [5] A. Edelman and H. Murakami, *Polynomial roots from companion matrix eigenvalues*, Mathematics of Computation 64 (1995) 763–776.
- [6] C47/R47 function reference. <https://47calc.com/documentation>
- [7] C47 developer wiki. <https://gitlab.com/rpncalculators/c43/-/wikis/home>

Colophon

Body and headings	TeX Gyre Heros, the metric-compatible Helvetica from TeX Live
Mathematics	New Computer Modern Sans Math, from TeX Live
Display face	JuliaMono, from juliaweb.org
Typeset with	lua _l atex, TeX Live 2025
Design language	c47appnote v1.1, docs/appnotes/sources/style
Charts	pgfplots, one hue and the grey magnitude ramp
Screen captures	the GTK simulator at --reset, upscaled 3× with a point filter
Written against	firmware 0.109.04.00.RC1

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